Walark Electronics · Fundrum Engineering

E4 Educator v2.0

Board Acceptance Test Procedure

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Property	Detail
Document title	E4 Educator v2.0 Board Acceptance Test Procedure
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Applies to	E4 Educator v2.0 (Enceladus Educator) — all production units
Test firmware	`E4_FACTORY_TEST v1.0.0 (E4-Educator-v2-Factory-Test.zip)`

1 Purpose and scope

This procedure defines the acceptance test to be performed on every E4 Educator v2.0 board before issue to a student or end user. The test exercises every hardware peripheral on the board using dedicated factory test firmware. A board that passes all tests is cleared for issue. A board that fails any test is quarantined for rework.

The test covers:

- All GPIO outputs (LEDs, piezo, DAC, TFT backlight)
- All GPIO inputs (buttons, LDR, MEMS microphone ADC, I2S)
- SPI bus peripherals (ILI9341 TFT display, XPT2046 touch controller, microSD card)
- I2C bus peripherals (AHT20 temperature/humidity, SSD1306 OLED if fitted)
- One-wire bus (DS18B20 temperature probe)
- WiFi radio (2.4GHz scan)
- TFT display quality (colour fills, sprite pipeline, touch zones)

> **Important**
> This test does NOT require a WiFi network connection, MQTT broker, or internet access. WiFi is tested in scan-only mode. All other tests are standalone.

2 Equipment required

Item	Specification / Notes
E4 Educator v2.0 board	Under test. ESP32 DevKit V1 socketed. TFT module fitted. DS18B20 probe connected to ONE_WIRE header.

microSD card	FAT32 formatted, 2–32GB, Class 4 minimum. Inserted before power-on.
USB-C cable	Data-capable (not charge-only). Connected to test PC running PlatformIO.
Test PC	Windows 10/11. PlatformIO installed in VS Code. CP2102 driver installed.
Factory test firmware	E4_FACTORY_TEST v1.0.0 — from production firmware package.
Headphones or speaker	Connected to DAC_AUDIO header (GPIO 25) or listening for piezo tone.
Oscilloscope (optional)	For DAC sine wave verification if audio output cannot be heard.
This test sheet	Printed. One copy per board. File in production record.

3 Loading the test firmware

1. Insert a FAT32 formatted microSD card into the TFT module SD slot.
2. Connect the E4 board to the test PC via USB-C.
3. Open the E4_FACTORY_TEST project folder in VS Code.
4. Confirm COM port in platformio.ini matches the board (monitor_port / upload_port).
5. Press Ctrl+Alt+U to build and upload the factory test firmware.
6. Open Serial Monitor (Ctrl+Alt+M). The firmware starts and reports its serial number status:

```
First test on this board (no NVS record):
  No NVS serial. Default: E4-000001  (send SN:XXXXX to set)

Retest after rework (NVS record already exists):
  NVS serial found: E4-000042  (send SN:XXXXX to override)

To set or override the serial number, type within 3 seconds:
  SN:E4-000042   (press Enter)

Use a unique incrementing serial number for each board.
If no command is entered the existing or default number is used.
```

Serial number persistence
On PASS the serial number is written to the ESP32 NVS flash in the dedicated 'factory' namespace. It survives power cycles, OTA firmware updates, and Settings.h factory reset operations — only running the factory test again can overwrite it. On retest after rework the stored serial number is loaded automatically so the operator does not need to re-enter it.

7. After 3 seconds the test sequence begins automatically.

4 Test sequence

The firmware runs tests in two phases. Automated tests complete without operator interaction. Interactive tests pause and prompt the operator on the TFT display.

4.1 Automated tests

These tests complete automatically within 15 seconds of power-on with no operator interaction:

No.	Test	Pass criterion	Typical result
01	I2C Bus scan	At least one device found on SDA/SCL bus	0x38 (AHT20)
02	AHT20 sensor	Temperature 5–50°C and humidity 5–99% returned	e.g. 21.4C 58.2%
03	DS18B20 probe	Device detected, temperature 5–80°C returned	e.g. 1 dev 22.1C
04	SD card	Mount, write, read-back verify, delete all pass	e.g. 32GB W:OK R:OK
05	SPI bus coexist	TFT renders correctly after SD transaction	TFT after SD
06	TFT sprite pipeline	8 sprite push operations complete without error	No flicker
07	WiFi radio scan	At least one 2.4GHz network detected	e.g. 8 nets
08	OLED SSD1306	0x3C present on I2C bus (PASS if fitted, INFO if not)	0x3C OK / Not fitted

4.2 Interactive tests

For each interactive test the TFT displays an instruction. The operator observes or interacts and then presses NEXT (pass) or ENT (fail). A long press of NEXT skips a test and marks it as skipped (not failed).

No.	Test	Operator action	Pass criterion
09	TFT backlight PWM	Watch backlight ramp from dim to bright and back	Smooth brightness change
10	TFT colour fills	Watch five full-screen colour fills: RED GREEN BLUE WHITE BLACK	All five colours distinct
11	LEDs RED/GRN/BLU	Watch LEDs: RED on/off, GREEN on/off, BLUE on/off, all on	Each LED illuminates correctly
12	Button NEXT	Press the NEXT button when prompted	TFT confirms detection
13	Button ENT	Press the ENT button when prompted	TFT confirms detection
14	Touch XPT2046	Tap the TFT screen. An orange dot appears at touch point.	Dot appears at touch point
15	LDR ADC1	Cover the LDR sensor with a finger when prompted, then release	ADC delta > 200 counts
16	Piezo buzzer	Listen for a scale of notes ascending then descending	Scale clearly audible
17	DAC audio GPIO25	Listen at DAC header or observe 440Hz on oscilloscope	Tone audible or measured
18	MEMS mic ICS43434	Clap or speak near the microphone when prompted	RMS > 50 counts detected

5 Interpreting results

5.1 TFT final screen

When all tests are complete the TFT shows a final result screen:

- Large green PASS — all tests passed. Board is cleared for issue.
- Large red FAIL — one or more tests failed. Board is quarantined for rework.
- The failed test names and detail strings are listed below the verdict.
- On PASS: a green confirmation line reads "✓ SN E4-XXXXXX stored in NVS" confirming the serial number has been written permanently to the board.
- On FAIL: the NVS is still written with the FAIL result so the board is identifiable on retest.
- GREEN LED blinks slowly on PASS. RED LED blinks slowly on FAIL.
- The bottom bar reads "Logged to SD + NVS | NEXT=restart".
- Pressing NEXT restarts the full test sequence for a retest after rework. The serial number is reloaded from NVS automatically.

5.2 Production records — NVS and SD card

The factory test writes records to two locations: the board's own NVS flash and the SD card.

NVS record (on the board)

Written to NVS namespace 'factory' after every test run (PASS or FAIL). Readable by curriculum firmware via FactoryInfo.h.

```
NVS namespace: 'factory'

Key            Value example         Notes
-------------- --------------------- ----------------------------------
serial         E4-000042             Board serial number
result         PASS                  PASS or FAIL
testFwVer      1.0.0                 Factory test firmware version
testDate       2026-05-15            Factory test build date
passCount      18                    Number of tests passed
failCount      0                     Number of tests failed
testedMs       47823                 Test session duration (ms)
```

The NVS record is read by curriculum firmware at startup via FactoryInfo.h:

```
#include "FactoryInfo.h"

FactoryInfo fi;
fi.load();
fi.print();               // Prints full record to Serial
fi.publishMQTT(mqtt);    // Publishes to e4/device/serial etc.

// MQTT topics published (all retained):
//   e4/device/serial      → "E4-000042"
//   e4/device/factoryTest → "PASS"
//   e4/device/testFw      → "1.0.0"
```

SD card records

Two files are written to the SD card at the end of every test run:

- FACTORY_LOG.CSV — cumulative log. One row per run. Survives multiple retest cycles on the same card.
- BOARD_E4-XXXXXX.TXT — full detail report for this board. Overwritten on retest.

```
FACTORY_LOG.CSV columns:
serial, fw, timestamp_ms, overall, passCount, failCount, failed_tests...

BOARD_E4-000042.TXT example:
E4 Educator v2.0 Factory Test Report
Board serial: E4-000042
Firmware:     1.0.0  2026-05-15
Test time:    47823ms
Overall:      PASS

No. Test                       Result Detail
--- ---                        ------ ------
01  I2C Bus scan               PASS   2 device(s): 0x38 0x3C
02  AHT20 sensor               PASS   21.4C 58.2%
...
18  MEMS mic ICS43434          PASS   RMS=2840 OK
```

5.3 Failure triage guide

Failed test	Likely cause	Rework action
I2C Bus scan — 0 devices	I2C pull-ups missing or SDA/SCL shorted	Check 4.7k pull-ups to 3V3 on SDA/SCL. Check solder joints at I2C connector.
AHT20 sensor — Not found	AHT20 not seated or wrong address	Reseat AHT20. Confirm address 0x38 in I2C scan results.
AHT20 — out of range values	Sensor fault or VCC not reaching device	Check 3V3 at AHT20 VCC pin. Replace AHT20 if 3V3 confirmed OK.
DS18B20 — No device	Probe not connected or 4.7k pull-up missing	Confirm probe plugged into ONE_WIRE header. Check 4.7k pull-up to 3V3 on GPIO 4.
SD card — Mount fail	Card not seated, SPI init order wrong, or wrong CS	Remove and reinsert card. Confirm FAT32 format. Check 10k pull-up on GPIO 15 (SD_CS).
TFT — Colour fills wrong	ILI9341 not initialising, SPI clock issue	Check TFT module seating. Confirm SPI_FREQUENCY=27000000 in build_flags.
TFT backlight — Always dim	BC847 transistor fault or GPIO 12 stuck	Measure base of BC847. Confirm GPIO 12 was LOW before LEDC setup. Replace BC847.
Touch — No detection	TOUCH_CS (GPIO 33) not connected or XPT2046 fault	Check TOUCH_CS line continuity. Confirm SPI_TOUCH_FREQUENCY=2500000.
LEDs — One not lighting	LED or series resistor fault	Measure LED anode/cathode. Check GPIO output voltage. Replace LED or resistor.
Button — Timeout	Button solder joint or INPUT_PULLUP not working	Measure button pin voltage at rest (should be 3V3). Reflow button solder joints.
LDR — Delta too low	LDR not fitted or divider resistor wrong	Check LDR seated in PCB. Confirm divider resistor value (typically 10k). Check ADC1 pin GPIO 39.

Piezo — Silent	Piezo not fitted or LEDC not reaching GPIO 27	Measure GPIO 27 with scope during test. Confirm piezo polarity. Replace if 3V3 square wave confirmed at pin.
MEMS mic — RMS low	I2S config wrong, BCLK missing, or mic orientation	ICS43434 is directional — confirm port is facing up/forward. Check GPIO 32 (BCLK) not sharing LEDC. Reflow mic solder joints.
WiFi — No networks	RF section fault or antenna issue	Board must be in an area with at least one 2.4GHz AP. If environment confirmed and still 0 networks: likely ESP32 antenna problem. Replace ESP32.

6 Board test sign-off sheet

Complete one copy of this sheet per board. File with production records.

Board serial number	E4 – _ _ _ _ _ _	**Test date**	/ /
Test firmware version	1.0.0	**Tested by**	

No.	Test	Pass criterion	Result	Notes / measured value
01	I2C Bus scan	At least 1 device found	☐	
02	AHT20 sensor	5–50°C and 5–99% RH	☐	
03	DS18B20 probe	Device present, 5–80°C	☐	
04	SD card	Mount/write/read/delete all pass	☐	
05	SPI bus coexist	TFT renders after SD transaction	☐	
06	TFT sprite	8 sprite pushes without error	☐	
07	WiFi radio	≥1 network detected	☐	
08	OLED SSD1306	0x3C detected (if fitted)	☐	
09	TFT backlight	Smooth ramp visible	☐	
10	TFT colour fills	5 colours distinct and correct	☐	
11	LEDs RGB	RED, GREEN, BLUE each illuminate	☐	
12	Button NEXT	Press detected within 8s	☐	
13	Button ENT	Press detected within 8s	☐	
14	Touch XPT2046	Tap point confirmed on screen	☐	
15	LDR ADC1	ADC delta > 200 on cover	☐	
16	Piezo buzzer	Scale of notes audible	☐	
17	DAC audio	440Hz sine audible / measured	☐	
18	MEMS mic	RMS > 50 on sound event	☐	

✓ PASS — CLEARED FOR ISSUE	✕ FAIL — QUARANTINE FOR REWORK
All tests passed. Signature: ________________	Failed tests circled above. Rework and retest.

Retest No.	Date	Rework performed	Result
Retest 1	/ /		☐ PASS ☐ FAIL
Retest 2	/ /		☐ PASS ☐ FAIL
Retest 3	/ /		☐ PASS ☐ FAIL

Record confirmation

- ☐ TFT shows '✓ SN E4-XXXXXX stored in NVS' on PASS result
- ☐ Serial monitor confirms '[NVS] Factory record written' message

☐ FACTORY_LOG.CSV updated on SD card
☐ BOARD_E4-XXXXXX.TXT file present on SD card
☐ SD card filed with this test sheet or digital record archived

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